

Call: HORIZON-MSCA-2026-PF-01

(MSCA Postdoctoral Fellowships 2026)

Topic: HORIZON-MSCA-2026-PF-01-01

Type of Action: HORIZON-TMA-MSCA-PF-EF

(HORIZON TMA MSCA Postdoctoral Fellowships - European Fellowships)

Proposal number: 101373105

Proposal acronym: FIELDWISE

Type of Model Grant Agreement: HORIZON Unit Grant

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Proposal ID 101373105
Acronym FIELDWISE

1 - General information

Call HORIZON-MSCA-2026-PF-01
Topic HORIZON-MSCA-2026-PF-01-01
Acronym FIELDWISE

Type of Action HORIZON-TMA-MSCA-PF-EF
Type of Model Grant Agreement HORIZON-AG-UN

Proposal title FIELDWISE — The Digital Agronomist: Physiologically Validated Plant-Based Water-Stress Prediction for Transferable Irrigation Decision Support in High-Value Horticulture

Scientific Areas

Panel	Sub-area of research	Descriptor
Environmental and Geosciences (ENV)	Applied Life Sciences and Non-Medical Biotechnology	Agriculture related to crop production, applied plant biology
Environmental and Geosciences (ENV)	Earth system science	Earth observations from space/remote sensing
Environmental and Geosciences (ENV)	Environment and society	Sustainable development and nature protection

IMPORTANT: Your proposal will be evaluated under the Scientific Panel of the 'Project descriptor' selected in the first line of the table above.

Free keywords Plant stress physiology; Water-deficit stress; Precision agriculture; Sentinel-2 remote sensing; Hyperspectral sensing; Predictive modelling; Data fusion; Digital agriculture; Sensing-scale transfer

Abstract

FIELDWISE addresses a central limitation in current crop water-stress prediction: the lack of physiologically validated models whose predictive validity is tested explicitly across years, environments and sensing scales before operational deployment. Building on a five-season (2022–2026) processing-tomato archive, the project will integrate physiological, soil-water, meteorological and proximal hyperspectral observations within a harmonised framework and derive Sentinel-2-compatible predictor definitions. An interpretable, uncertainty-aware probabilistic model will be developed under leakage-safe unseen-year validation and frozen before any prospective-field outcome is observed. Its transferability will then be evaluated over two commercial-field seasons using actual Sentinel-2, soil-water and meteorological inputs, while coordinated plant-physiological measurements provide independent evidence of realised stress. FIELDWISE will quantify not only predictive discrimination, but also calibration, uncertainty, false-alarm behaviour, warning lead time and the magnitude of the experimental-to-commercial-field transfer gap before any limited recalibration. Only scientifically validated functions will be incorporated into the MVP, which will then be tested by a commercial farm-management information system (FMIS) software company within its network of farmers to assess operational usability, interoperability and practical relevance under operational farming conditions. The project therefore aims to establish a transferable and auditable framework for precision-irrigation decision support, distinguishing components that remain robust across sensing scales and production environments from those requiring crop-specific recalibration and revalidation.

Has this proposal (or a very similar one) been submitted in the past 2 years in response to a call for proposals under any EU programme, including the current call? ☐ Yes ☒ No

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Declarations

We declare to have the explicit consent of all applicants on their participation and on the content of this proposal. ☒

We confirm that the information contained in this proposal is correct and complete and that none of the project activities have started before the proposal was submitted (unless explicitly authorised in the call conditions). ☒

We declare: ☒

- to be fully compliant with the eligibility criteria set out in the call
- not to be subject to any exclusion grounds under the EU Financial Regulation
- and to have the financial and operational capacity to carry out the proposed project.

We acknowledge that all communication will be made through the Funding & Tenders Portal electronic exchange system and that access and use of this system is subject to the Funding & Tenders Portal Terms and Conditions. ☒

We have read, understood and accepted the Funding & Tenders Portal Terms & Conditions and Privacy Statement that set out the conditions of use of the Portal and the scope, purposes, retention periods, etc. for the processing of personal data of all data subjects whose data we communicate for the purpose of the application, evaluation, award and subsequent management of our grant, prizes and contracts (including financial transactions and audits). ☒

We declare that the proposal complies with ethical principles (including the highest standards of research integrity as set out in the ALLEA European Code of Conduct for Research Integrity, as well as applicable international and national law, including the Charter of Fundamental Rights of the European Union and the European Convention on Human Rights and its Supplementary Protocols. Appropriate procedures, policies and structures are in place to foster responsible research practices, to prevent questionable research practices and research misconduct, and to handle allegations of breaches of the principles and standards in the Code of Conduct. ☒

We declare that the proposal has an exclusive focus on civil applications (activities intended to be used in military application or aiming to serve military purposes cannot be funded). If the project involves dual-use items in the sense of Regulation 2021/821, or other items for which authorisation is required, we confirm that we will comply with the applicable regulatory framework (e.g. obtain export/import licences before these items are used). ☒

We confirm that the activities proposed do not: ☒

- aim at human cloning for reproductive purposes;
- intend to modify the genetic heritage of human beings which could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed);
- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer;
- lead to the destruction of human embryos (for example, for obtaining stem cells).

These activities are excluded from funding.

We confirm that for activities carried out outside the Union, the same activities would have been allowed in at least one EU Member State. ☒

The coordinator is only responsible for the information relating to their own organisation. Each applicant remains responsible for the information declared for their organisation. If the proposal is retained for EU funding, they will all be required to sign a declaration of honour.

False statements or incorrect information may lead to administrative sanctions under the EU Financial Regulation.

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2 - Participants

List of participating organisations

#	Participating Organisation Legal Name	Country	Role
1	HUN-REN Agrartudományi Kutatóközpont	Hungary	Coordinator
2	AGROVIR Üzletviteli Tanácsadó Korlátolt Felelősségű Társaság	Hungary	Associated

end of list

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Proposal ID **101373105**
Acronym **FIELDWISE**
Short name **HUN-REN Agrártudományi Kutatóközpont**

Organisation data

PIC	Legal name		
866599553	HUN-REN Agrartudomanyi Kutatokozpont		
Short name	HUN-REN Agrártudományi Kutatóközpont		
Street	Brunszvik u. 2.	Legal person	Yes
Town	Martonvásár	Public body	No
Postcode	2462	International organisation	No
Country	Hungary		

Legal Status

Research organisation	Secondary or higher education
Yes	No
Non-profit	Academic
Yes	Yes
Mid-cap	Gender equality plan
No	Yes
International European research organisation	
No	

SME data

SME self-declared status	SME self-assessment	SME validation
02/04/2026 No	- Unknown	- No

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Short name **HUN-REN Agrártudományi Kutatóközpont**

Departments carrying out the proposed work

Department 1

Department name

Plant Physiology and Metabolomics

☐ not applicable

☐ Same as proposing organisation's address

Street

Brunszvik utca 2

Town

Martonvásár

Postcode

2462

Country

Hungary

Links with other participants

Type of link	Participant

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Proposal ID **101373105**
Acronym **FIELDWISE**
Short name **HUN-REN Agrártudományi Kutatóközpont**

Supervisor

Title

Prof

Gender

☐ Woman

☒ Man

☐ Non Binary

First name

Tibor

Last name

Janda

E-Mail

janda.tibor@atk.hun-ren.hu

Position in org.

Head of Department

Department

Department of Plant Physiology and Metabolomics

☒ Same as proposing organisation's address

Street

Brunszvik u. 2.

Town

Martonvásár

Post code

2462

Country

Hungary

Website

https://atk.hun-ren.hu

Phone

+3622569500

Phone 2

+3622569570

Other contact persons

First Name	Last Name	E-mail	Phone
Andrea	Kelényi	kelenyi.andrea@atk.hu	+36 22569500
Katalin	Parádi	paradi.katalin@agrar.mta.hu	+36 22569565
Jenő	Kontschán	atk.palyazat@atk.hu	+36 22569565

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Short name **HUN-REN Agrártudományi Kutatóközpont**

Researcher

The name and e-mail of the Researcher and Supervisor are read-only in the administrative form, only additional details can be edited here. To give access rights and contact details of contact persons, please go back to Participants page of the submission wizard and save the changes.

First Name	Rositsa	Last Name	Cholakova
E-Mail	rositsa.cho@abv.bg	Phone	+359885752888
Last Name at Birth	Cholakova	Gender	☒ Woman ☐ Man ☐ Non binary
Title	Dr.	Country of residence	Bulgaria
Nationality	Bulgaria	Nationality 2	
Country of birth	Bulgaria	Date of Birth (DD/MM/YYYY)	07/06/1989
Place of Birth	Plovdiv		

Contact address

Current organisation name	Bulgarian Agricultural Academy - MVCRI
Current Department/Faculty/ Institute/Laboratory name	Plant physiology

☐ Same as proposing organisation's address

Street	Brezovsko shose 32		
Postal code	4003	Town	Plovdiv
Country	Bulgaria	Phone2 / Mobile	+359885752888
ORCID	0009-0008-3761-3106	Researcher ID	
Other ID	Scopus Author ID: 57209853646		

Qualifications

Applicants who have successfully defended their doctoral thesis but who have not yet formally been awarded the doctoral degree will also be considered as postdoctoral researchers and will be considered eligible to apply. A medical doctor degree will be accepted only when it corresponds to a doctoral degree or if the researcher can demonstrate his/her appointment in a position that requires doctoral equivalency (e.g. professorship appointment). Medical doctor degrees corresponding to basic medical training as defined in Annex V of Directive 2005/36/EC will not be considered a doctoral degree.

Doctorate Date of (expected) award (DD/MM/YYYY)	14/10/2020
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Short name **HUN-REN Agrártudományi Kutatóközpont**

Place of activity/place of residence (previous 5 years - most recent one first)

Indicate the period(s) and the country/countries in which you have legally resided and/or had your main activity (work, studies, etc) during the last 5 years up until the deadline for the submission of the proposal.

Please fill in this section without gaps

Period from (DD/MM/YYYY)	Period to (DD/MM/YYYY)	Duration (days)	Country
05/06/2026	09/09/2026	96	Bulgaria
09/03/2026	04/06/2026	87	Hungary
09/01/2021	08/03/2026	1,884	Bulgaria
Total duration		2,067	

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Short name **HUN-REN Agrártudományi Kutatóközpont**

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other Role	☐
If yes, please specify: (Maximum number of characters allowed: 50)	

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Short name HUN-REN Agrártudományi Kutatóközpont

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Janda et al., 2025 – Scientific Reports: water-deficit physiology; chlorophyll fluorescence, stomatal conductance, SPAD. Szalai et al., 2025 – Plant Physiology and Biochemistry: whole-plant abiotic-stress physiology. Hollós et al., 2026 – Geoscientific Model Development: ML, leave-one-year-out validation, independent testing, SHAP. Birinyi et al., 2024 – Remote Sensing: EO-based maize condition mapping. Li et al., 2026 – Nature Plants: plant signalling and stress-response physiology.

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Plants under stress	HUN-REN ATK contributes relevant experience from Shoot-root signalling in plants under stress, combining physiology, gene expression and metabolomics, and from European projects including LANDRACES (MSCA IF 752453), AI4SoilHealth, COUSIN and TUDI. Together these demonstrate strong capacity in plant-stress research, digital and data-driven agriculture, international collaboration, researcher training and Horizon Europe project implementation.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure or equipment	Short description (Max 300 characters)
Advanced plant phenotyping platform, Photosynthesis	Walk-in phenotyping chambers with multispectral fluorescence, RGB, VNIR/SWIR hyperspectral, thermal and 3D imaging; LI-COR 6400 and Imaging PAM; Waters UPLC/HPLC, LECO/Shimadzu GC-MS, CAMAG HPTLC and Shimadzu UV/VIS for physiological and metabolomic plant-stress analysis.

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Short name **Agrovir**

Organisation data

PIC	Legal name		
891807622	AGROVIR Uzletviteli Tanacsado Korlatolt Felelossegu Tarsasag		
Short name	Agrovir		
Street	Kinizsi utca 30	Legal person	Yes
Town	Budaors	Public body	No
Postcode	2040	International organisation	No
Country	Hungary		

Legal Status

Research organisation	Secondary or higher education
No	No
Non-profit	Academic
No	No
Mid-cap	Gender equality plan
No	No
International European research organisation	
No	

SME data

SME self-declared status	SME self-assessment	SME validation
- Unknown	- Unknown	- No

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Acronym **FIELDWISE**
Short name **Agrovir**

Departments carrying out the proposed work

Department 1

Department name

☐ not applicable

☐ Same as proposing organisation's address

Street

Town

Postcode

Country

Links with other participants

Type of link	Participant

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Proposal ID **101373105**
Acronym **FIELDWISE**
Short name **Agrovir**

Role of participating organisation in the project

- Project management ☐
- Communication, dissemination and engagement ☒
- Provision of research and technology infrastructure ☐
- Co-definition of research and market needs ☒
- Civil society representative ☐
- Policy maker or regulator, incl. standardisation body ☐
- Research performer ☐
- Technology developer ☐
- Testing/validation of approaches and ideas ☒
- Prototyping and demonstration ☒
- IPR management incl. technology transfer ☐
- Public procurer of results ☐
- Private buyer of results ☐
- Finance provider (public or private) ☐
- Education and training ☐
- Contributions from the social sciences or/and the humanities ☐
- Other Role ☐
- If yes, please specify: (Maximum number of characters allowed: 50)

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Short name **Agrovir**

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Software	<i>AgroVIR develops digital farm-management software that can integrate with other agricultural software and data sources and support the monitoring of field operations. Its solutions help manage farm activities, records, machinery, inputs, costs and operational data, supporting coordinated monitoring and data-driven farm management.</i>

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
<i>Precision Decision Centre, SMART-GAZDA, Bonafarm G</i>	<i>AgroVIR has led and participated in digital-agriculture projects including the Precision Decision Centre, SMART-GAZDA, Bonafarm Group FMIS implementation and AgroVINO. These activities covered integration of software, sensors and machinery, farm-data management, precision decision support, fleet tracking, SAP interoperability, field monitoring, traceability and deployment of digital solutions in commercial farming environments.</i>

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure or equipment	Short description (Max 300 characters)
<i>AgroVIR digital farm-management infrastructure</i>	<i>AgroVIR provides cloud-based FMIS infrastructure, API and data-integration interfaces, and the AgroVIR Works mobile application. Together, these support integration of farm, machinery, weather and operational data, field-level monitoring, digital work records, satellite-map use, navigation and excha</i>

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3 - Budget

Is the Researcher eligible for family allowance? ☒ Yes ☐ No

Duration of the Fellowship	Country in which fellowship will take place
24	Hungary
Non-academic placement duration (1 to 6 months)	Country in which non-academic placement will take place
6	Hungary

	Country Coefficient	Number of Months	Contributions for recruited researchers			Institutional contributions		Total
			Living Allowance	Mobility Allowance	Family Allowance	Research, training and networking costs	Management and indirect costs	
Fellowship	0.787	24	119,938.8	17,040	15,840	24,000	15,600	192,418.8
Placement in non-academic sector	0.787	6	29,984.7	4,260	3,960	6,000	3,900	48,104.7
Total			149,923.5	21,300	19,800	30,000	0	240,523.5

The Partner Organisation does not sign the Grant Agreement and does not directly claim costs from the action. The entire EU contribution is transferred to the Host organisation located in a Member State or Associated Country.

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4 - Ethics & security

Ethics Issues Table

1. Human embryonic stem cells and human embryos		Page
Does this activity involve Human Embryonic Stem Cells (hESCs)?	☐ Yes ☒ No	
Does this activity involve the use of human embryos?	☐ Yes ☒ No	
2. Humans		Page
Does this activity involve human participants?	☐ Yes ☒ No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	☐ Yes ☒ No	
Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014)? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	☐ Yes ☒ No	
3. Human Cells / Tissues (not covered by section 1)		Page
Does this activity involve the use of human cells or tissues?	☐ Yes ☒ No	
4. Personal data		Page
Does this activity involve processing of personal data?	☐ Yes ☒ No	
Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)?	☐ Yes ☒ No	
Is it planned to export personal data from the EU to non-EU countries?	☐ Yes ☒ No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country?	☐ Yes ☒ No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	☐ Yes ☒ No	
5. Animals		Page
Does your research involve animals?	☐ Yes ☒ No	
6. Non-EU countries		Page
Will some of the activities be carried out in non-EU countries?	☐ Yes ☒ No	
In case non-EU countries are involved, do the activities undertaken in these countries raise potential ethics issues?	☐ Yes ☒ No	
Is it planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	☐ Yes ☒ No	
Is it planned to import any material from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4.	☐ Yes ☒ No	
Is it planned to export any material from the EU to non-EU countries?	☐ Yes ☒ No	
Does this activity involves low and/or lower-middle income countries? (if yes, detail the benefit-sharing actions planned in the self-assessment)	☐ Yes ☒ No	
Could the situation in the country put the individuals taking part in the activity at risk?	☐ Yes ☒ No	
7. Environment, Health and Safety		Page
Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants (during the implementation of the activity or further to the use of the results, as a possible impact)?	☐ Yes ☒ No	
Does this activity deal with endangered fauna and/or flora / protected areas?	☐ Yes ☒ No	

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Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity (during the implementation of the activity or further to the use of the results, as a possible impact)?		☐ Yes	☒ No	
8. Artificial intelligence				Page
Does this activity involve the development, deployment and/or use of Artificial Intelligence-based systems?		☒ Yes	☐ No	6
9. Other ethics issues				Page
Are there any other ethics issues that should be taken into consideration?		☐ Yes	☒ No	

I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines How to Complete your Ethics Self-Assessment ☒

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Ethics Self-Assessment

Ethical dimension of the objectives, methodology and likely impact
FIELDWISE is an agricultural research and innovation project focused on plant water-stress physiology, Earth observation, IoT-enabled sensing, predictive modelling and irrigation decision support. The research does not involve human biological material, medical research, clinical interventions, animals, vulnerable groups or special-category personal data. The limited human dimension concerns professional end-user engagement during the requirements, usability and operational evaluation of the DrR Digital Agronomist decision-support prototype. Growers and professional users may provide structured feedback on usability, interpretation of outputs, missing-data behaviour, interoperability and practical relevance. Participation will be voluntary. Participants will receive clear information about the purpose of the activity, the type of information collected and its intended use, and will be free to withdraw without disadvantage. Only data necessary for the evaluation will be collected. Personal identifiers will not be required for the scientific objectives. Where participant or farm-related information is recorded, it will be minimised and coded or anonymised wherever possible. Commercially sensitive farm information will not be disclosed in publications; scientific reporting will use aggregated, coded or anonymised results. FIELDWISE also develops and uses machine-learning-based prediction models for plant water-stress assessment and irrigation decision support. The AI component uses agronomic, physiological, meteorological, IoT and Earth-observation data. It is not designed to make decisions concerning health, employment, access to services, legal rights or other high-impact human contexts. The system is designed as decision support rather than autonomous control: stress probability, uncertainty, temporal trends and irrigation-attention information are presented to the farmer or agronomist, who retains responsibility for management decisions. Ethical AI principles are incorporated into the methodology through transparent documentation of model purpose and limitations, leakage-safe validation, calibration and uncertainty assessment, model-card documentation, interpretable modelling where appropriate, and prospective testing in an independent commercial-field environment. Performance deterioration, uncertainty and false alarms will be reported rather than concealed. These safeguards reduce risks of inappropriate reliance on model outputs and support transparent, accountable and human-supervised use of the resulting technology.

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Compliance with ethical principles and relevant legislations
All FIELDWISE activities will comply with Horizon Europe ethical principles, applicable EU legislation, Hungarian national requirements and the internal research-integrity and data-protection procedures of HUN-REN ATK and the participating organisations. Any involvement of growers or other professional end users in requirements gathering or operational evaluation will follow the principles of voluntary participation, informed consent, transparency, data minimisation and the right to withdraw. No vulnerable populations will be specifically recruited and no physical, medical, behavioural or clinical intervention is involved. Where required by the host or participating institution, the relevant institutional ethics or data-protection procedure will be completed before participant-related activities begin. Processing of personal data will comply with the General Data Protection Regulation (EU) 2016/679 (GDPR) and applicable national data-protection legislation. Only data that are necessary for the defined project purpose will be processed. Direct identifiers will be avoided wherever possible; data will be coded, pseudonymised or anonymised as appropriate, access will be limited to authorised project personnel, and data will be stored using secure institutional infrastructure. Personal data will not be transferred to non-EU countries. Commercial farm and partner data will be handled according to agreed access, confidentiality and intellectual-property arrangements. Publications and publicly released datasets will contain no identifiable personal information or confidential commercial data. The development and use of machine-learning components will follow principles of human oversight, transparency, robustness, accountability and appropriate data governance, taking account of applicable EU requirements concerning trustworthy AI. The FIELDWISE AI models support agricultural decision-making and do not replace the farmer or agronomist as the final decision-maker. Model performance, calibration, uncertainty, transferability, intended use and limitations will be documented, including through a model card. The model will be validated on unseen years and prospectively tested before validated functions are incorporated into the operational DrR interface. Research outputs will additionally follow Horizon Europe requirements on research integrity, FAIR data management and open science under the principle “as open as possible, as closed as necessary.” Protection of personal data, commercially sensitive information and potentially exploitable intellectual property will take precedence where unrestricted disclosure would be inappropriate.

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Security Issues Table

1. EU Classified Information (EUCI) ²		Page
Does this activity involve information and/or materials requiring protection against unauthorised disclosure (EU)?	☐ Yes ☒ No	
Does this activity involve participants from non-EU countries which need to have access to EUCI?	☐ Yes ☒ No	
2. Misuse		Page
Does this activity have the potential for misuse of results?	☐ Yes ☒ No	
3. Other Security Issues		Page
Does this activity involve information and/or materials subject to national security restrictions? If yes, please specify: (Maximum number of characters allowed: 1000)	☐ Yes ☒ No	
Are there any other security issues that should be taken into consideration? If yes, please specify: (Maximum number of characters allowed: 1000)	☐ Yes ☒ No	

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Security Self-Assessment

FIELDWISE is a civilian agricultural research project and does not involve classified information, security-sensitive infrastructure, restricted technologies or activities presenting a foreseeable risk of misuse. Open Copernicus Earth-observation data and research agronomic data will be used; any operational or commercially sensitive information provided by the producer or AgroVIR will be protected through institutional access controls, data minimisation, coding or aggregation, and appropriate confidentiality agreements. Proportionate access, storage and cybersecurity measures will also apply to the DrR prototype and its source code. No additional security information or security screening is therefore considered necessary.
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² According to the Commission Decision (EU, Euratom) 2015/444 of 13 March 2015 on the security rules for protecting EU classified information, “European Union classified information (EUCI) means any information or material designated by an EU security classification, the unauthorised disclosure of which could cause varying degrees of prejudice to the interests of the European Union or of one or more of the Member States”.

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5 - Other questions

Information on the Researcher (future fellow)

Were you in the last 3 years in compulsory national service? ☐ Yes ☒ No

Did you spend time, in the last 3 years, on procedures for obtaining refugee status (according to the 1951 Geneva Refugee Convention and the 1967 Protocol) in a Member State or Associated Country to Horizon Europe? ☐ Yes ☒ No

Are you a national of a Member State or Associated Country? ☒ Yes ☐ No

Country: Bulgaria

Other Questions

Do you wish to participate to the ERA Postdoctoral Fellowships and thus increase your chances of being funded? If you don't reply or if you maintain 'No' as your answer, be aware that your proposal will not benefit from an extra chance of being funded under the ERA Postdoctoral Fellowships in case it cannot be funded under the MSCA Postdoctoral Fellowships (MSCA-PF). A dedicated budget from the Work Programme "Widening participation and strengthening the European Research Area" has been set aside to fund those proposals. European Postdoctoral Fellowships proposals where the host organisation is located in an eligible widening country (according to the PIC number) but were not funded under MSCA-PF call due to a lack of budget, can be automatically duplicated into the ERA Postdoctoral Fellowships call. Eligible proposals will be duplicated into the ERA Postdoctoral Fellowships call only if you express your wish to do so. Your decision not to participate in the ERA Postdoctoral Fellowships will not affect your chances of being funded directly under the MSCA-PF call. ☒ Yes ☐ No

Are the research and training activities proposed addressing topics related to nuclear sciences and technology, as outlined in the Guide for applicants for this call and the Euratom Research and Training Programme (ERTP)? ☐ Yes ☒ No

Answer "Yes" **ONLY IF** all three conditions below are fulfilled:

1. The proposal's research area is covered by the ERTF.
2. The host organisation (and, if applicable, the Associated Partner for the additional Placement period) is/are established in a Member State or Associated Country to the ERTF.
3. The researcher is a national or a long-term resident of a Member State or Associated Country to the ERTF.

For communication purposes only, the European Commission REA asks for permission to publish the name of the researcher (future fellow) should the proposal be retained for funding. Does the researcher (future fellow) give this permission? ☒ Yes ☐ No

Some national and regional public research funding authorities run schemes to fund MSCA applicants that score highly in the MSCA evaluation but which cannot be funded by the MSCA due to their limited budget. In case this proposal could not be selected for funding by the MSCA, do the researcher and supervisor consent to the European Commission disclosing to such authorities the results of its evaluation (score and ranking range) together with their names and contact details, non-confidential proposal title and abstract, proposal acronym, and host organisation? ☒ Yes ☐ No

Part B-1

1. Excellence

1.1 *Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)*

Problem and overarching aim. European agriculture faces increasing water constraints: approximately 30% of EU territory experiences seasonal water scarcity in an average year, while climate change is expected to intensify drought frequency, severity and seasonal freshwater fluctuations¹. High-value horticultural irrigation often relies on delayed visual symptoms, isolated sensor thresholds or retrospective crop-performance records, risking mistimed or unnecessary irrigation. FIELDWISE asks: **can a physiologically grounded plant water-stress signal remain valid across seasons, environments and sensing scales, survive prospective validation under real commercial-field conditions, and be translated into crop-configurable irrigation decision support?** Processing tomato is the **development and validation crop**, supported by an unusually rich five-season (2022–2026) archive of physiological, soil, meteorological and proximal-spectral observations, enabling rigorous testing beyond a short-term fellowship-generated dataset. Unchanged transfer to other crops is not assumed: FIELDWISE will distinguish reusable methodological components from biological and calibration elements requiring crop-specific recalibration and revalidation. An end-of-project non-academic placement completes inter-sectoral translation.

Pertinence. Three linked scientific gaps motivate the objectives: (i) insufficient physiological grounding of water-stress indicators; (ii) limited evidence of predictive validity across unseen years, environments and sensing scales; and (iii) weak translation into operational irrigation decision support. Spectral and environmental variables frequently serve as stress proxies without demonstrated consistency with plant physiological responses². FIELDWISE centres prediction on physiology and tests transferability prospectively rather than inferring it from retrospective accuracy. These objectives also advance the fellow from plant physiologist and agronomist toward an independent **Agricultural Data Scientist** integrating plant physiology, Earth observation, IoT-enabled sensing, predictive modelling and decision-support development. Scientific and career objectives thus reinforce each other and align directly with MSCA Postdoctoral Fellowships' interdisciplinary and inter-sectoral profile.

The four objectives, and why they are measurable and verifiable. **O1 — Harmonise the 2022–2026 MATE archive and establish the water-stress reference and validation protocol.** The five-season processing-tomato archive will be harmonised across physiological, soil-water, meteorological and hyperspectral observations collected under contrasting irrigation treatments. These treatments provide a known water-deficit context, with irrigation requirements historically derived using AquaCrop-based crop-water-demand calculations, as established in previous MATE processing-tomato research^{3,4}. Historical hyperspectral measurements will be converted, where technically possible, to Sentinel-2-compatible predictor definitions and temporally aligned with the other observations. Outputs are the harmonised database and predictor-availability map (**D1.1, M2**) and the stress-reference, feature-selection and physiological-validation protocol (**D1.2, M3**). **O2 — Develop an interpretable and uncertainty-aware probabilistic water-stress model.** Under HUN-REN ATK supervision, with Prof. András Jung/ELTE supporting hyperspectral-to-Sentinel harmonisation, the model will integrate Sentinel-compatible spectral indices reconstructed from historical hyperspectral observations with soil-water and meteorological variables. Irrigation treatment provides the imposed water-deficit context, while physiological measurements such as chlorophyll fluorescence and stomatal conductance provide independent evidence of whether that deficit produced an actual plant stress response. The primary model output will be water-stress probability and operational stress status. The model, benchmark, predictor definitions and model card will be frozen at **M3 (D1.4)**. **O3 — Prospectively validate the frozen model under unseen commercial-field conditions over two processing-tomato seasons.** Validation will be conducted in a commercial-scale field large enough for valid Sentinel-2 observation and containing contrasting controlled irrigation regimes, providing non-stressed and water-deficit conditions. The model will receive actual Sentinel-2-derived indices together with soil-water and meteorological observations. Coordinated physiological measurements will independently verify the plant response. Correct detections, missed detections, false alarms, calibration and warning lead time will be quantified. Outputs are the

1 <https://www.eea.europa.eu>

2 Gerhards M. et al. (2019), Challenges and Future Perspectives of Multi-/Hyperspectral Thermal Infrared Remote Sensing for Crop Water-Stress Detection: A Review, *Remote Sensing* 11, 1240.

3 Takács S. et al. (2020), Influence of Water Stress Levels on the Yield and Lycopene Content of Tomato, *Water* 12, 2165.

4 Takács S. et al. (2021), Performance Evaluation of AquaCrop Model in Processing Tomato Biomass, Fruit Yield and Water Stress Indicator Modelling, *Water* 13, 3587.

prospective validation dataset (**D2.1, M20**) and transferability and uncertainty report (**D2.2, M21**). **O4 — Translate validated outputs into the DrR web MVP and assess their operational relevance.** FIELDWISE builds on the fellow's pre-existing desktop DrR research prototype. Krumatic will provide remunerated software-engineering support for development of the web MVP, while AgroVIR will contribute operational requirements, user feedback and final non-academic evaluation. Only scientifically validated model outputs will enter the operational prototype. Outputs are the web MVP and readiness dossier (**D3.1–D3.2, M24**) and the operational/interoperability assessment and roadmap (**D4.1–D4.2, M30**).

Measurability and verifiability. Model development within the historical archive will use blocked, leakage-safe unseen-year validation. The stress-reference definition, predictors, feature-selection rules, evaluation metrics and permitted recalibration strategy will be pre-specified in D1.2. The frozen model will subsequently be tested prospectively over two unseen commercial-field seasons. Performance will be evaluated using balanced accuracy, F1 and ROC-AUC for stress detection; reliability and Brier score for probabilistic calibration; and lead time and false-alarm rate for operational value. Regression metrics will be used only if the archive supports a defensible continuous stress-severity reference. Freezing the model and predictor definitions at M3 prevents retrospective optimisation against prospective outcomes and makes the central hypothesis directly falsifiable: whether water-stress relationships learned from controlled experimental observations remain valid when corresponding predictors are obtained from actual Sentinel-2 observations under commercial-field conditions.

Realistically achievable. FIELDWISE combines a 24-month MSCA Postdoctoral Fellowship at HUN-REN ATK with a six-month AgroVIR non-academic placement. The existing five-season MATE processing-tomato archive and controlled irrigation experiments support feasibility. Previous MATE research established AquaCrop-based full-to-deficit irrigation treatments and demonstrated the value and limitations of modelling water stress from irrigation and crop-water-demand information alone. FIELDWISE links this imposed deficit context to physiological plant response and transferable spectral predictors. Prospective validation covers one commercial environment, two growing seasons and controlled irrigation contrasts; DrR builds on an existing prototype. In-action validation remains limited to processing tomato; a crop-transfer specification and subsequent post-project replication and knowledge transfer through MVCRI address additional crops.

Beyond the state of the art, and the extent of the ambition. FIELDWISE challenges four assumptions: vegetation indices themselves represent water stress; proximal and satellite observations are interchangeable; internal validation demonstrates field transferability; and one crop's model transfers unchanged to another. Its ambition is a physiologically validated, uncertainty-aware transfer framework, not a new machine-learning algorithm, testing whether relationships learned from controlled experiments remain valid across unseen years, sensing scales and commercial fields. Three citable advances are: (1) a probabilistic processing-tomato water-stress model documenting calibration, uncertainty and blocked unseen-year performance; (2) a prospective experimental-to-commercial-field transfer gap quantified with a model frozen before deployment; and (3) a specification separating reusable modelling components from sensing-scale and crop-specific elements requiring recalibration. The new knowledge quantifies **whether, how far and under what conditions predictive validity survives transfer from controlled experiments to operational satellite-based monitoring.**

1.2 Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices).

Overall methodology: concepts, models and assumptions. Four linked criteria evaluate water-stress prediction: physiological validity, predictive performance, transferability and management relevance. Operational predictors comprise spectral, soil-water and meteorological variables. Historical MATE controlled irrigation treatments and AquaCrop-informed crop-water requirements establish the imposed deficit context; physiological measurements independently indicate whether it produced actual plant stress. Before model fitting, D1.2 will specify how these complementary sources define the stress reference.

Stage 1 (WP1) harmonises the 2022–2026 MATE archive into a QC-controlled common-variable matrix with explicit missingness and availability maps. Historical plots are too small for defensible plot-level Sentinel-2 retrieval; handheld hyperspectral measurements will therefore be transformed, where technically possible, into Sentinel-2-compatible predictors before model freeze. A 12-month ELTE research secondment under Prof. András Jung, distributed across the 24-month fellowship, supports predictor harmonisation and subsequent prospective Sentinel-2 validation/EO-transfer assessment. Blocked, leakage-safe unseen-year validation tests temporal transfer; a parsimonious uncertainty-aware model is compared with a nonlinear benchmark. Predictors, validation rules and the primary model are frozen at M3, before use of any prospective-field outcome.

Stage 2 (WP2) prospectively tests environmental and sensing-scale transfer over two processing-tomato seasons in an independent commercial field. Controlled irrigation creates well-watered, deficit-irrigated and unirrigated conditions in treatment areas permitting valid Sentinel-2 observation. The frozen model uses actual Sentinel-2 indices, soil-water and meteorological observations; coordinated physiology independently verifies realised plant stress. Primary performance precedes any recalibration; subsequent limited recalibration is secondary and pre-specified. The transfer gap quantifies predictive validity retained from controlled experiments to genuine satellite-scale field monitoring (**D2.1–D2.2, M20–M21**). Crop transfer is addressed separately, distinguishing reusable components from those requiring crop-specific recalibration and revalidation.

Stage 3 (WP3–WP4) operationalises only validated functions. DrR – Digital Agronomist enters FIELDWISE as the fellow’s pre-existing, independently developed research prototype. FIELDWISE validates and extends selected scientific functionalities, while Krumatic provides the corresponding software-engineering implementation and AgroVIR evaluates operational usability and interoperability.

Methodological challenges and mitigation. (i) A year-by-variable availability matrix and only defensible common predictors manage inter-annual archive heterogeneity; (ii) pre-model spectral harmonisation and prospective scale-transfer testing link hyperspectral and satellite observations without assuming interchangeability; (iii) blocked unseen-year validation avoids optimistic random-split generalisation; (iv) a model frozen before deployment tests commercial-field transfer, reporting performance before recalibration; (v) sufficiently large treatment areas and valid interior observations reduce Sentinel mixed-pixel risk; and (vi) parsimony, uncertainty assessment, lead time and false-alarm rate control over-fitting. If clouds or acquisition timing prevent usable Sentinel-2 observations at critical validation dates, targeted UAV flights may supply supplementary high-resolution observations, subject to platform availability and flight conditions. These support interpretation but do not replace Sentinel-2 in the primary transferability assessment. A predefined site-contingency plan protects two-season field validation.

Integration of methods and disciplines. FIELDWISE combines plant physiology, agronomy, Earth observation, predictive modelling and software engineering. HUN-REN ATK supplies plant-stress physiology/modelling; MATE the multi-season controlled-irrigation archive and processing-tomato expertise; Prof. András Jung/ELTE hyperspectral-to-Sentinel harmonisation, EO methodology and sensing-scale transfer through a distributed 12-month secondment; and Krumatic/AgroVIR software-engineering and operational farm-management perspectives. The fellow integrates these disciplines in a common workflow: the biological reference constrains model interpretation, the protocol constrains development, and the validated predictor specification governs operational field inputs.

Gender dimension and other diversity aspects. A sex/gender dimension is not relevant to the biological research content, which concerns processing-tomato water-stress responses and involves no human or animal biological data. Diversity becomes relevant only in end-user interaction: usability feedback will consider differences in farm context and digital literacy. Any grower feedback will follow informed-consent, withdrawal, data-minimisation and coded/anonymised-processing principles.

Open science practices. FIELDWISE follows the principle “as open as possible, as closed as necessary.” Stress-reference rules, predictor definitions, evaluation metrics and acceptance criteria will be pre-specified in D1.2, while D1.4 includes a model card documenting intended use, validation, uncertainty and limitations. Data and metadata will be managed according to FAIR principles, Sentinel-2 inputs originate from open Copernicus data, and both successful and unsuccessful transfer results will be reported. At least two open-access peer-reviewed papers are targeted. A Data Management Plan will be completed within the first six months, covering data standards, storage, access, sharing and reuse. Analysis code, metadata and non-sensitive derived outputs will be shared where institutional, commercial and IP rights permit; potentially protectable results will be assessed before public disclosure

1.3 Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

Supervisory architecture and qualifications. FIELDWISE is supported by complementary expertise in plant-stress physiology, predictive modelling and Earth observation. Primary supervision at the recruiting organisation, HUN-REN ATK, is provided by **Prof. Tibor Janda**, Head of the Plant Physiology and Metabolomics Department at the Agricultural Institute. His expertise in plant physiology, abiotic stress, including drought, and stress metabolomics anchors interpretation of plant water-stress responses, the physiological-validation framework and

the measurement programme. **Dr Roland Hollós** supports the predictive-modelling and machine-learning strand⁵, including interpretable modelling, uncertainty analysis and leakage-safe validation. **Prof. András Jung (ELTE, Institute of Cartography and Geoinformatics)** provides specialist co-supervision in field spectroscopy, hyperspectral imaging, geoinformatics, satellite Earth observation and sensing-scale transfer⁶. MATE contributes the five-season controlled-irrigation processing-tomato archive and agronomic expertise through **Dr Sándor Takács**, including AquaCrop-based irrigation and crop-water-stress methodology, complemented by precision-agriculture expertise from **Prof. Gábor Milics**. The host also provides demonstrated European-project experience: ATK coordinated the MSCA Individual Fellowship **LANDRACES (752453)** and participates in Horizon Europe projects including **COUSIN, AI4SoilHealth and TUDI**.

Structured supervision and governance. Supervision comprises regular one-to-one meetings with Prof. Janda, dedicated modelling/EO technical sessions with specialists, monthly written progress monitoring and quarterly Career Development Plan reviews. The **Project Steering Group**—fellow, primary supervisor, specialist co-supervisor/modelling expert and relevant partner contacts—reviews protocol/model freeze, prospective-field implementation, transfer-validation completion, MVP readiness and the non-academic placement. Pre-defined decision responsibilities and a two-step conflict-resolution route ensure continuity while protecting the fellow's growing scientific independence.

Planned training activities and ELTE secondment. Training is embedded in implementation. ATK deepens physiological water-stress interpretation, phenotyping, experimental design, predictive modelling, calibration and uncertainty analysis. A 12-month ELTE research secondment, distributed across the 24-month fellowship, provides advanced supervised training in hyperspectral-to-Sentinel harmonisation, EO processing, geoinformatics and sensing-scale transfer, supporting early Sentinel-compatible predictor construction and later analysis of actual prospective-field Sentinel-2 observations. These activities advance the fellow toward an independent **Agricultural Data Scientist** at the physiology–EO–predictive decision-support interface. Transferable skills include research integrity, open science/FAIR practice, responsible generative-AI use, grant writing, IP/exploitation, project management and science communication. Guest lectures and practical activities at MATE and, where feasible, Agricultural University–Plovdiv build on previous university teaching experience. The **Career Development Plan (D5.1, M3)** consolidates the programme and is reviewed throughout the action.

Two-way transfer of knowledge. FIELDWISE connects internationally recognised plant-stress physiology, predictive-modelling, EO and precision-irrigation expertise. HUN-REN ATK contributes advanced abiotic-stress physiology, phenotyping and modelling; ELTE established hyperspectral-sensing, geoinformatics and satellite-EO expertise. The fellow contributes drought-stress physiology, remotely monitored plant-stress responses, field physiological phenotyping and interpretation under contrasting water availability, strengthening stress biology's translation into irrigation decision support. Joint supervision, the distributed ELTE secondment, shared MATE-archive analysis, prospective field validation, teaching and co-authored publications reinforce exchange.

Rationale and added value of the non-academic placement. The six-month AgroVIR placement (M25–M30) is an integral inter-sectoral component of FIELDWISE, providing competences not fully available academically: DrR web-MVP operational evaluation, FMIS requirements engineering, interoperability assessment, user-oriented testing and exploitation planning. AgroVIR supplies requirements feedback during the fellowship; the placement adds intensive exposure to production agricultural software and decision-support workflows. Structured technical reviews assess usability, data-flow requirements, missing-data behaviour, interoperability and the pathway toward future FMIS integration. Commercially sensitive evaluation uses protected, coded or appropriately aggregated data. Confidentiality, DrR background access, ownership/use of new results and exploitation arrangements will be defined in advance under the Grant Agreement and relevant institutional/contractual arrangements.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

A profile matched to the project's interdisciplinary chain. Dr. Rositsa Cholakova (publishing before 2020 as Rositsa Cholakova-Bimbalova) holds BSc and MSc degrees in Plant Protection and a PhD in Plant Physiology (2020, Agricultural University of Plovdiv). She was Assistant Professor in Plant Physiology and Plant Stress Physiology in Plovdiv (2015–2020) and is currently Chief Assistant Professor at the Maritsa Vegetable Crops Research Institute, where her work addresses abiotic stress and physiological phenotyping. Her expertise includes drought-stress physiology, chlorophyll fluorescence, stomatal responses, pigment status, leaf water status and field-

5 Hollós R. et al. (2026), Meta-modelling of carbon fluxes from crop and grassland multi-model outputs, *Geoscientific Model Development* 19, 4385–4438.

6 Jung A. et al. (2010), White-reference based post-correction method for multi-source spectral libraries, *PFG* 2010(5), 363–369.

based stress assessment. This background is central to FIELDWISE because it enables remotely sensed and model-derived signals to be interpreted against actual plant physiological responses rather than treated as stress proxies by default. **Data-science and Earth-observation competences already in place.** The fellow has deliberately expanded this physiological background toward agricultural data science and remote sensing. Her experience includes agricultural data work at **Kleffmann Group Bulgaria**, formal upskill training in **Artificial Intelligence, Data Science, Machine Learning and Python**, and practical use of R, QGIS, Google Earth Engine, SNAP and satellite/UAV data. She has undertaken a COST PANGEOS Short-Term Scientific Mission and Erasmus+ training at MATE, an ELTE traineeship in geoinformatics/remote sensing, and further international mobility at the University of Novi Sad. She holds an EU A1/A3 drone-pilot qualification, is holding Environmental Engineering Master's degree, and has independently developed the pre-existing **DrR – Digital Agronomist** desktop research prototype. Five years of university teaching in Bulgarian and English provide an additional foundation in communication, mentoring and knowledge transfer. **Fit and development potential.** The fellow already combines the biological, agronomic and technical competences required to lead FIELDWISE, while the fellowship addresses the next step in her progression: consolidating advanced predictive modelling, uncertainty analysis, leakage-safe validation and transfer across years, sensing scales and commercial-field conditions. The 12-month distributed ELTE secondment and specialist supervision at HUN-REN ATK are therefore designed to deepen an existing interdisciplinary trajectory rather than create it from zero. Following a full-time-parenting career break (2021–2024), she has re-entered research with a focused transition toward an independent **Agricultural Data Scientist working at the interface of plant physiology, Earth observation and irrigation decision support.**

2. Impact

2.1 *Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development*

FIELDWISE is designed to consolidate the fellow's transition from plant-stress physiologist and agronomist to an **independent Agricultural Data Scientist working at the interface of plant physiology, Earth observation, predictive modelling and irrigation decision support.** The fellowship builds on competences already established in physiology, agronomy, EO and data analysis and adds the advanced capabilities required for scientific independence: leakage-safe predictive modelling, uncertainty and calibration, sensing-scale transfer, operational EO workflows, research-to-software translation and inter-sectoral exploitation.

Two realistic post-fellowship trajectories anchor this development. The first is an independent research/data-science role in European academia, research organisations or agri-technology, supported by a demonstrable record spanning physiological validation, predictive modelling, Sentinel-based field transfer and industrial evaluation. The second is development of an independent research and innovation line in Bulgaria, extending the FIELDWISE methodology to further irrigated high-value crops and transferring the acquired competences through MVCRI. Progress is tracked through the Career Development Plan and career KPI **K11**.

Measure	Skills and employability mechanism	Evidence
Career Development Plan	Co-developed with the supervisors; translates career objectives into technical training, publications, teaching, networking, leadership and open-science actions, with progress reviewed at major project transitions.	D5.1 (M3); reviews M3/M12/M21/M24/M27; K11
ATK scientific and modelling training	Advanced plant-stress interpretation, experimental design, interpretable predictive modelling, leakage-safe validation, calibration and uncertainty analysis.	Supervision records; model card; D1.4
12-month distributed ELTE secondment	Advanced hyperspectral-to-Sentinel harmonisation, geoinformatics, EO processing and sensing-scale transfer, applied first to historical predictor construction and later to real Sentinel-2 field observations.	Secondment activities; CDP; EO/transfer outputs
Transferable-skills portfolio	Research integrity, FAIR/open science, responsible AI use, grant writing, IP/exploitation, project management and science communication, exercised directly through FIELDWISE.	CDP-tracked training and project outputs

Measure	Skills and employability mechanism	Evidence
Non-academic placement	Six months at AgroVIR (M25–M30) developing competence in FMIS requirements, interoperability, user-oriented evaluation and exploitation planning.	D4.1–D4.2 (M30)
Scientific visibility and leadership	Co-authored publications, conference presentations, seminars and teaching activities strengthen international visibility, communication, network building and future supervision capacity.	≥2 manuscripts; conference/teaching outputs

By completion, the fellow will have taken one research concept through **biological definition, historical-data harmonisation, predictive modelling, unseen-year validation, prospective satellite-based field validation, software translation and industrial evaluation**. This combination is uncommon within a single researcher profile and strengthens employability across academic plant/EO research, agricultural data science and agri-technology. The 12-month ELTE secondment adds sustained inter-institutional experience and EO specialisation, while the AgroVIR placement provides direct non-academic experience. Together with the FIELDWISE publication, software and transfer-validation record, these measures provide concrete evidence of progression toward research independence rather than training participation alone.

2.2 *Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities*

FIELDWISE uses a targeted strategy proportionate to a single-fellow action, progressing from **scientific dissemination** → **practitioner communication** → **operational and entrepreneurial exploitation**. Target groups are research communities, growers and advisors, FMIS/agri-software and technology-transfer actors, students and the wider public.

Strand	Target groups	Measures and outputs	Evidence/timing
Dissemination	Plant physiology, crop modelling, RS/EO, precision agriculture; students	≥2 peer-reviewed manuscripts, possible third paper; model card; hyperspectral-to-Sentinel/transfer methodology; publication of prospective transfer performance; EGU/ECPA presentations; seminars and teaching. FAIR release of methods, metadata, code and non-sensitive outputs where rights permit.	D1.4 M3; D2.1 M20; D2.2 M21
Communication	Growers, advisors and public in HU/BG	Practitioner demonstrations of stress probability, uncertainty and irrigation-attention outputs; public-facing articles; farmer/end-user workshop; institutional and public-engagement activities. Claims are restricted to measured project evidence.	Demonstrations, workshops, reach indicators
Exploitation	FMIS/agri-software sector, growers, technology-transfer actors	Maturation of the fellow's pre-existing DrR desktop prototype into a validated web MVP . FIELDWISE provides validated scientific/EO functionality; Krumatic provides remunerated software engineering; AgroVIR provides requirements and operational evaluation. Post-MSCA routes include FMIS integration, MVCRI crop extension and HUN-REN/HUNRENTech-supported licensing or researcher-led start-up/spin-off exploration.	D3.1–D3.2 M24; D4.1–D4.2 M30
IP and knowledge management	Fellow, beneficiary and contributing organisations	Separate rights layers for DrR background / FIELDWISE-generated results / institutional datasets / contracted code . A dated DrR declaration establishes the pre-action baseline. Ownership/licensing of the new web software and source code is defined through the relevant HUN-REN ATK–Krumatic agreement; AgroVIR receives only access required for evaluation.	Pre-action agreements; maintained IP inventory

Exploitation pathway. DrR enters FIELDWISE as the fellow's pre-existing, independently developed research prototype. Only scientifically validated functions are transferred into the web MVP. At M24 the result is a validated MVP, followed by independent AgroVIR evaluation in M25–M30 before any claim of FMIS readiness or wider commercial deployment. FIELDWISE therefore delivers four exploitable result families: the validated stress-prediction workflow, the hyperspectral-to-Sentinel transfer methodology, the matured DrR web MVP, and the AgroVIR-informed FMIS-integration specification.

Reach and uptake are tracked through concrete indicators: at least two conference presentations, at least two submitted manuscripts, practitioner demonstrations, institutional seminars, an end-user workshop, public-engagement activities, AgroVIR operational assessment and monitoring of repository use after release.

2.3. The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

FIELDWISE is a **24-month MSCA Postdoctoral Fellowship followed by a six-month non-academic placement**. Its impact is bounded to what the action can demonstrate directly: whether physiologically grounded water-stress relationships remain valid across unseen years, sensing scales and two prospective commercial-field seasons, and whether validated results can support operational decision-making.

Dimension	Contribution	Magnitude and importance
Scientific	Leakage-safe modelling links physiological evidence with spectral, soil-water and meteorological predictors; a frozen model is tested across unseen years and an independent commercial field.	FIELDWISE determines whether, how far and under which conditions predictive validity survives transfer from experimental data to operational Sentinel-based monitoring . Outputs include a documented model, quantified transfer performance and a specification separating reusable from crop-/scale-specific components.
Agricultural & environmental	The workflow provides stress probability, uncertainty and irrigation-attention information independently checked against plant physiology across two seasons.	The direct value is more defensible identification of when and where water stress requires attention . Warning lead time, false alarms, uncertainty and water–yield–quality relationships are measured prospectively; no numerical water-saving benefit is claimed in advance.
Technological & economic	The fellow's pre-existing DrR desktop prototype is matured into a validated web MVP; Kromatic implements the software layer and AgroVIR evaluates usability and interoperability.	The pathway is auditable: pre-existing prototype → validated web MVP → independent operational assessment → post-MSCA FMIS/exploitation roadmap . This reduces the scientific and operational uncertainty required for later integration, licensing or researcher-led innovation.
European & societal	HUN-REN ATK, ELTE, MATE, Bulgarian research and agri-software actors are linked in a Hungary–Bulgaria knowledge-transfer chain.	Cross-border circulation of physiology, EO, data science and software know-how creates reusable European capacity, while MVCRI provides a realistic route for later extension to further high-value irrigated crops.

Impact is measured through **project-controlled evidence**: discrimination and calibration, uncertainty, false alarms, warning lead time and prospective transfer performance, all pre-specified in D1.2. Primary performance is reported before any limited recalibration so that the research-to-field transfer gap remains visible. Delivery is tracked through **K1–K14**, covering archive harmonisation and stress-reference definition, frozen model and transfer specification, prospective validation, web-MVP readiness and external evaluation, post-MSCA exploitation planning and Career Development Plan implementation. The funded magnitude is intentionally realistic: **one development crop, one large commercial validation environment across two seasons, one validated web MVP and one external industrial evaluation**. The wider value lies in the reusable transfer methodology and the defined post-MSCA scientific, institutional and entrepreneurial pathways.

3. Quality and Efficiency of the Implementation

3.1 *Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages*

The fellow leads the scientific work at the sole beneficiary, HUN-REN ATK (ATK), over 24 months, followed by the AgroVIR placement (M25–M30). Five WPs connect a frozen historical-data model, two prospective tomato seasons, evidence-limited software translation and career development. ELTE secondment blocks are M1–M3, M10–M12 and M16–M21 (12 months); they are locations of work within the allocated effort. MATE supplies archive/irrigation expertise; the producer hosts field trials; Krumatic supplies remunerated engineering.

WP1 — Data harmonisation, stress definition & model development | M1–M21 | 5.0 PM.

Lead: fellow/ATK; MATE and ELTE support. **T1.1** (M1–M2): audit and harmonise all five archive seasons. **T1.2** (M1–M3): harmonise predictors and lock the physiological reference, folds, metrics, sampling and recalibration rules before fitting. **T1.3** (M2–M3): build the pipeline; after protocol lock, fit and freeze the model/benchmark. **T1.4** (M10–M12; M16–M21): synthesise WP2 transfer diagnostics without changing the primary model.

D1.1 archive/QC map (M2); D1.2 protocol and D1.4 model/card (M3); D1.5 transfer specification (M21). K1–K4: five seasons audited; one locked protocol; reproducible, leakage-safe held-out-year results; one reusable/crop-specific transfer specification. Gates: MS1/MS3.

WP2 — Prospective field validation | M1–M21 | 9.0 PM.

Lead: fellow/ATK; producer, MATE and ELTE support. **T2.1** (M1–M4): agree access/backup, replication, irrigation contrasts and valid Sentinel-pixel geometry; test sensors before observations. **T2.2** (M4–M12) and **T2.3** (M13–M20): run seasons 1 and 2 using the unchanged model, actual Sentinel-2 inputs and independent physiology; record irrigation, yield and quality. **T2.4** (M20–M21): report primary performance before separately evaluated recalibration.

D2.1 dataset/prediction log (M20); D2.2 transfer/uncertainty report (M21). K5: two seasons; all scheduled observation opportunities accounted for, with valid coverage and biological sample size reported. K6: season-level and pooled discrimination, calibration, false alarms, lead time and transfer gap, with uncertainty or non-estimability explained. Gates: MS1–MS3.

WP3 — DrR web MVP and readiness | M1–M24 | 6.0 PM.

Lead: fellow/ATK; Krumatic engineering; AgroVIR requirements. **T3.1** (M1–M3): agree background/IP, user workflows, interfaces and acceptance tests. **T3.2** (M4–M20): build ingestion, provenance and uncertainty displays using test data/research-only functions. **T3.3** (M21–M24): integrate MS3-supported functions and test scientific consistency and invalid-input behaviour. **T3.4** (M22–M24): package the release and placement protocol.

D3.1 web MVP and D3.2 readiness dossier (M24). K7: one MVP, all agreed critical tests pass and zero unresolved critical defects in the released scope. K8: source access, rights, evidence limits, guidance and evaluation scope documented. Gate: MS4.

WP4 — AgroVIR placement | M25–M30 | 5.4 PM.

Lead: fellow with AgroVIR operational supervision; ATK oversight. **T4.1** (M25–M27): assess data requirements, output consistency, missing/out-of-domain behaviour, usability, interoperability and decision relevance. **T4.2** (M28–M30): retest priority issues and record limitations. **T4.3** (M27–M30): define the post-MSCA integration/exploitation route using approved test/coded data.

D4.1 external assessment and D4.2 roadmap (M30). K9: all six dimensions assessed and priority issues dispositioned. K10: roadmap states owners, IP/access, technical dependencies, actors, resources and decision gates. Gates: MS5/MS6.

WP5 — Training, open science, dissemination and management | M1–M30 | 4.6 PM.

Lead: fellow/ATK; supervisors and host support services contribute. **T5.1:** implement the CDP, quarterly reviews, technical/transferable skills and knowledge exchange. **T5.2:** implement data/access rules from M1 and maintain the DMP. **T5.3:** run dissemination/communication/exploitation from M1; document the initial strategy at M3. **T5.4:** monitor progress/risks monthly and manage agreements and reporting (all M1–M30).

D5.1 CDP (M3); D5.2 DMP (M6; review M21/M30); D5.3 final dissemination/exploitation plan (M30; update M24). K11: CDP reviews, four technical competence strands and 12 secondment months evidenced. K12: all released assets have metadata and access decisions. K13: ≥ 2 manuscripts submitted, ≥ 2 conference presentations and ≥ 1 end-user workshop by M30. K14: each result family has an owner, audience, route and indicators. Required

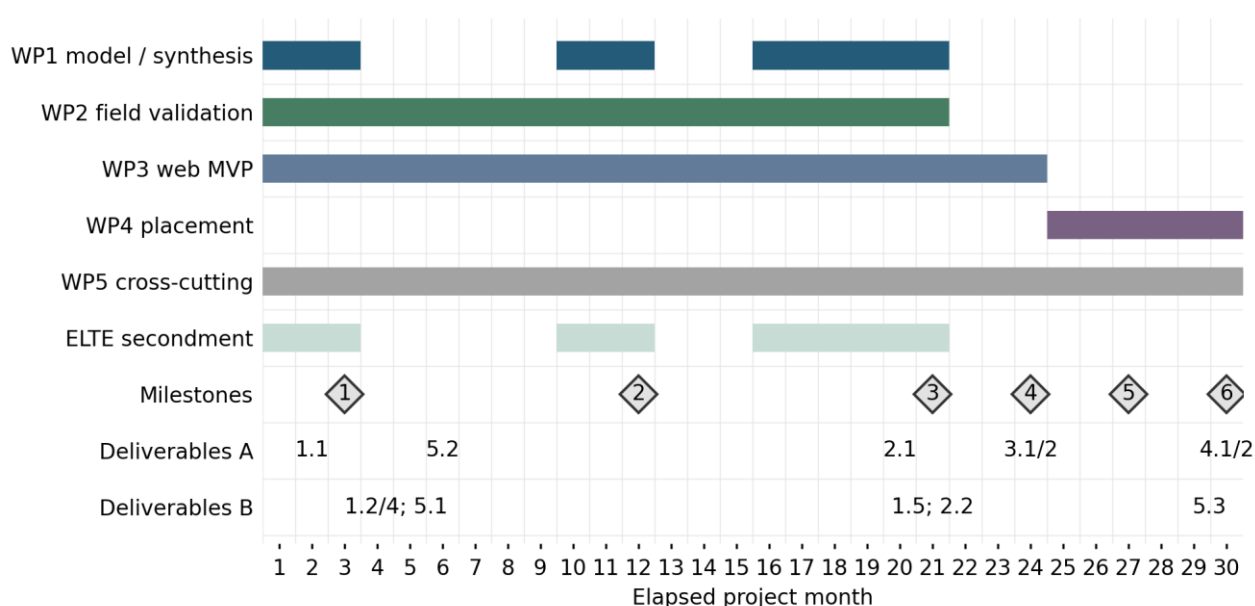
administrative outputs: mobility declaration within 20 days of research-training start, updated as needed; end-of-training questionnaire and follow-up two years later.

Decision gates and dependencies. MS1 (M3): protocol lock precedes fitting and model freeze; field launch plan/CDP agreed. MS2 (M12): season 1 QC and logistics review, retaining the primary model. MS3 (M21): two-season transfer evidence and specification accepted; permitted MVP functions decided. MS4 (M24): critical tests and placement scope accepted. MS5 (M27): midpoint assessment/corrections agreed. MS6 (M30): external assessment and roadmaps completed. The fellow and ATK approve scientific gates with specialist input; AgroVIR co-reviews MS4–MS6.

Critical path: MS1 model freeze → WP2 prospective evidence → MS3 permitted functions → MS4 release → WP4 evaluation. Field preparation and software architecture start earlier; they do not require WP1 to finish. The crop calendar must place MS1 before season 1 and the second harvest by M20. WP effort totals 30.0 fellow PM: 24.0 in M1–M24 and 6.0 in M25–M30 (WP4 5.4; WP5 0.6). Training within scientific tasks is counted once. Partner support is additional to these fellow PM.

Risk control. Monthly monitoring and gate reviews record triggers, owners and corrective decisions. Poor transfer is a reportable scientific result; it does not authorise unsupported advice.

Risk; likelihood/impact	Trigger response and responsible actors
R1 WP1 M/H	Archive/sample-size or M3-freeze failure: audit at M1–M2; narrow common predictors/reference before deployment; never use prospective outcomes to repair training. Owner: fellow/ATK.
R2 WP2 M/H	Field/calendar failure: agree access, replication, valid pixels and backup by M3; test sensors before observations. Switch comparable site before the crop window; downgrade claims if two seasons fail. Owner: fellow/ATK/producer.
R3 WP2 M/H	Sensor failure/weak stress contrast: continuous QC, spares/manual checks; physiology verifies stress. Report weak contrasts and missing metrics. Owner: fellow/ATK/producer.
R4 WP1–2 M/H	Clouds/mixed pixels/EO access: agree ELTE access and pixel rules; log gaps. UAV is supplementary; inadequate Sentinel coverage restricts satellite-transfer claims. Owner: fellow/ELTE.
R5 WP2–3 M/H	Poor transfer: report unchanged-model performance first; evaluate recalibration separately. Limit MVP/placement to supported functions. Owner: fellow/ATK/AgroVIR.
R6 WP3–5 M/M	Engineering/IP/workload delay: agree scope and source rights before development; review releases monthly, protect campaigns and narrow noncritical features. Owner: fellow/ATK; Kromatic/AgroVIR.



3.2 *Quality and capacity of the host institutions and participating organisations, including hosting arrangements*

The sole beneficiary and recruiting organisation is HUN-REN ATK. Detailed infrastructure and facilities are provided in Part B-2 Section 5. Each participating organisation contributes a complementary capacity required for implementation of the work plan.

Organisation	Capacity and arrangements
HUN-REN ATK – beneficiary, recruiting organisation and academic host	HUN-REN research organisation, Martonvásár, with the fellowship anchored in the Agricultural Institute. Prof. Tibor Janda is the primary supervisor , providing scientific supervision in plant-stress physiology, phenotyping and abiotic-stress research. Dr Roland Hollós supports the predictive-modelling/ML strand. Where available and formally assigned early-stage female researchers will participate in field measurements under supervision , providing practical research-training opportunities and supporting gender-balanced participation. ATK provides the scientific environment, infrastructure, workspace, administration and project support required throughout the action.
ELTE – 12-month research secondment host	The Institute of Cartography and Geoinformatics provides expertise in hyperspectral sensing, geoinformatics, Earth observation and sensing-scale transfer. Prof. András Jung is the co-supervisor and secondment supervisor . The 12-month research secondment is distributed across the 24-month fellowship to support early hyperspectral-to-Sentinel harmonisation and prospective Sentinel-2 transfer assessment. Dr Zsófia Varga supports targeted UAV acquisition where required and feasible; UAV observations remain supplementary and do not replace Sentinel-2 in the primary transferability assessment. Where available, female students will participate in EO/geospatial processing and measurement activities under supervision , strengthening practical training and gender-balanced participation.
MATE – historical-data and irrigation-science contributor	Provides the 2022–2026 controlled-irrigation processing-tomato archive , including soil-water, physiological, spectral, meteorological and agronomic observations. Dr Sándor Takács contributes processing-tomato irrigation and AquaCrop expertise, while Prof. Gábor Milics contributes precision-agriculture expertise. Data access and permitted reuse are documented before analysis and exploitation.
Krumatic EOOD – remunerated software-engineering support	Krumatic provides paid specialist software-engineering services for implementation of the DrR web MVP from the fellow's pre-existing desktop research prototype. Scientific specifications, model validation and interpretation remain with the fellow/HUN-REN ATK. The development agreement defines software deliverables, source-code delivery and relevant ownership, licensing and access rights.
AgroVIR Kft. – non-academic placement host	Hungarian FMIS company. AgroVIR provides operational requirements and feedback during the fellowship and hosts the M25–M30 non-academic placement , where the validated DrR web MVP is evaluated for usability, decision relevance, data requirements and interoperability. Proprietary customer-data access is not a project dependency.
Commercial farmer/producer – prospective field-trial host and validation environment	An independent Hungarian farmer/producer, irrespective of gender , provides the commercial field for two prospective seasons , implements the agreed irrigation treatments and carries out routine crop management according to the field protocol. The project installs soil-water and meteorological monitoring and conducts coordinated physiological and EO observations. Experimental design, stress-reference definition, validation, analysis and scientific interpretation remain with the fellow/HUN-REN ATK.
MVCRI, Plovdiv – post-MSCA continuation route	Prof. Ganeva supports the institutional route for post-MSCA continuation at MVCRI. The institute provides a realistic pathway for crop-specific recalibration, further validation in other high-value irrigated vegetable crops, and transfer of FIELDWISE knowledge to Bulgarian production conditions. MVCRI has no core in-action validation role during the project.

Part B-2

4. CV of the researcher

Dr Rositsa Cholakova (publishing before 2020 as Rositsa Cholakova-Bimbalova) is a plant physiologist and agronomist combining plant-stress physiology, physiological phenotyping, Earth observation and agricultural data science. She is currently Chief Assistant Professor at the Maritsa Vegetable Crops Research Institute (MVCRI), Plovdiv, Bulgaria.

Education and research qualifications

- PhD in Plant Physiology, Agricultural University of Plovdiv, 2020. Thesis: Research of the reaction of maize (*Zea mays* L.) to low temperatures and the effectiveness of subsequent foliar fertilisation.
- MSc in Plant Protection (Agronomy), Agricultural University of Plovdiv, 2014.
- BSc in Plant Protection (Agronomy), Agricultural University of Plovdiv, 2012.
- Second MSc studies in Environmental Engineering (2025–2026), strengthening environmental-systems and geospatial competences.

Professional experience

- Chief Assistant Professor, MVCRI, Plovdiv: abiotic-stress physiology, vegetable-crop phenotyping, field experimentation and non-destructive stress assessment.
- Assistant Professor in Plant Physiology / Plant Stress Physiology, Agricultural University of Plovdiv (2015–2020), including university teaching in Bulgarian and English.
- Agricultural data work at Kleffmann Group Bulgaria, adding commercial agricultural-data experience.
- Independent developer of the pre-existing DrR – Digital Agronomist desktop research prototype.

Methods and technical competences

Plant physiology: chlorophyll fluorescence (Fv/Fm), stomatal conductance/resistance, pigment/chlorophyll status, relative water content, leaf/canopy temperature, gas exchange and field stress assessment. EO/geospatial: Sentinel-1/2, hyperspectral/multispectral data, QGIS, SNAP, Google Earth Engine and UAV observations. Data science: Python, R, interpretable ML workflows and formal training in AI, Data Science, Machine Learning and Python. EU A1/A3 drone-pilot qualification.

International mobility and advanced training

- COST Action CA22136 PANGEOS Short-Term Scientific Mission at MATE – 10 days (2025).
- Erasmus+ staff mobility - RGB/hyperspectral training at MATE – 14 days (2025).
- ELTE traineeship in geoinformatics and remote sensing - 3 months (2026).
- Erasmus+ staff mobility at the University of Novi Sad, Faculty of Agriculture – 7 days (2026).

Selected relevant research outputs

- Selected peer-reviewed publications
1. **Cholakova, R., Petrov, V., & Vasilev, A. (2019).** Photosynthetic performance of young maize (*Zea mays* L.) plants exposed to chilling stress can be improved by the application of protein hydrolysates. *Acta Agrobotanica*, **72**(2), 1769.
 2. **Kamenova, I., Dimitrov, P., Ganeva, D., & Cholakova, R. (2025).** Remote sensing monitoring of watermelon under different water regimes: Methodology and preliminary results. In *Proceedings of SES 2025 – Space, Ecology, Safety*, Sofia, Bulgaria.
 3. **Cholakova, R., & Vasilev, A. (2017).** Effect of chilling stress on the photosynthetic performance of young plants from two maize (*Zea mays* L.) hybrids. *CBU International Conference Proceedings*, 1118–1123.
 4. **Cholakova, R., Koleva, L., & Vasilev, A. (2018).** Effects of a biostimulant and a mineral fertilizer on the antioxidative defence system of chilling-exposed maize plants. *Agricultural Sciences*, 33–39.
 5. **Cholakova, R. (2015).** Influence of low temperatures on the growth and macronutrient content in young maize plants. *Scientific Works, Agricultural University – Plovdiv*, **LIX**(2), 87–94.

6. **Chipilski, R., Kyosev, B., & Cholakova, R. (2020).** Changes in the growth and antioxidant enzyme activity of seedlings originating from wheat seeds subjected to accelerated ageing test. *Agricultural Science and Technology*, **12**(3), 216–220. <https://doi.org/10.15547/ast.2020.03.034>
 7. **Dragov, R., Taneva, K., Hadzhiivanova, B., Videva, M., Todorova, B., & Cholakova, R. (2025).** Genetic nature of quantitative traits in durum wheat. *Scientific Papers. Series A. Agronomy*, No. 1, 320–326. Web of Science: WOS:001598862000037.
 8. **Dragov, R., Taneva, K., Hadzhiivanova, B., Videva, M., Todorova, B., & Cholakova, R. (2026).** Graphical diallel analysis for quantitative traits in durum wheat. *Scientific Papers. Series A. Agronomy*, No. 2, 382–391. Web of Science: WOS:001658972500001.
- Manuscripts under review

Kamenova, I., Dimitrov, P., Ganeva, D., Cholakova, R., Filchev, L., Ivanov, I., Tringovska, I., Ganeva, D., & Grozeva, S. (2026). *Estimation of Evapotranspiration in Tomato Fields from Multispectral and Thermal UAV Remote Sensing Data. Under review.*

Takács, S., Pék, Z., Daood, H. G., Ganeva, D., Grozeva, S., Tringovska, I., & Cholakova, R. (2026). *Proximal Hyperspectral Signatures of Water Stress in Processing Tomato: Physiological and Soil-Water Validation. Under review.*

Career break

Full-time parenting career break: 34 months (2021–2024). The fellow subsequently re-entered research with a focused transition toward agricultural data science, Earth observation and irrigation decision support.

5. Capacity of the Participating Organisation(s)

5.1 Template table: Overview of Participating Organisations

Organisation role	PIC	Legal Entity Short Name	Academic organisation (Y/N)	Country	Name of Supervisor
Beneficiary	866599553	HUN-REN ATK	Y	Hungary	Prof. Tibor Janda
Associated partner for secondment	999896468	ELTE	Y	Hungary	Prof. András Jung
Associated partner for non-academic placement	891807622	AgroVIR	N	Hungary	Miklós Maróti

Only formal MSCA participating organisations are listed. MATE contributes the historical archive and scientific expertise; Krumatic provides remunerated software-engineering support; the commercial farmer/producer hosts and implements the agreed field-trial irrigation regimes; and MVCRI is a post-MSCA continuation route.

5.2 Template table: Capacity of the Participating Organisations

Beneficiary
HUN-REN Centre for Agricultural Research (HUN-REN ATK) – Hungary

General description	HUN-REN ATK is a public agricultural research organisation in Martonvásár. FIELDWISE is anchored in the Agricultural Institute. As sole beneficiary and employing organisation, ATK provides scientific integration, workspace, administration and responsibility for implementation throughout the fellowship.
Role and profile of supervisor	Prof. Tibor Janda, DSc, Head of Plant Physiology and Metabolomics, is the primary supervisor. His expertise in plant physiology, abiotic stress and metabolomics anchors the physiological-validation framework. Dr Roland Hollós provides complementary predictive-modelling/ML mentoring.
Key research facilities, infrastructure and equipment	ATK provides plant-physiology/metabolomics laboratories, field and controlled-environment facilities, and a large Fitotron complex with automated phenotyping capacity for stress experiments, physiological measurements and data analysis.
Previous and current involvement in EU-funded research and training programmes/actions/projects	HUN-REN ATK coordinated the H2020 MSCA Individual Fellowship LANDRACES (GA 752453) and has participated in Horizon Europe projects including AI4SoilHealth (GA 101086179) and COUSIN (GA 101135314).

Associated partner for secondment	
Eötvös Loránd University (ELTE) – Hungary	
General description	The 12-month distributed secondment at the Institute of Cartography Science and Geoinformatics supports hyperspectral-to-Sentinel harmonisation, EO processing, geoinformatics and sensing-scale transfer across early model construction and prospective validation.
Role and profile of supervisor	Prof. András Jung, Professor and Deputy Director, is co-supervisor and secondment supervisor; his expertise spans hyperspectral/field spectroscopy, high-resolution remote sensing and multisensor fusion.
Key research facilities, infrastructure and equipment	Geoinformatics Laboratory; remote-sensing/geospatial computing; field-spectroscopy and UAV survey capacity.
Previous and current involvement in EU-funded research and training programmes/actions/projects	ELTE coordinates the Horizon Europe QUEST Twinning project (GA 101156088); Prof. Jung also has COST Action experience relevant to international training and knowledge transfer.

Associated partner for non-academic placement	
AGROVIR Üzletviteli Tanácsadó Kft. (AgroVIR Kft.) – Hungary	
General description	AgroVIR is a Hungarian non-academic FMIS company headquartered in Budaörs, with more than a decade of multi-country implementation experience. It hosts the M25–M30 placement for operational evaluation of the validated DrR web MVP.

Role and profile of supervisor	Miklós Maróti, Managing Director and agricultural engineer, is the placement supervisor. His experience in agricultural production, consultancy and farm digitalisation provides the required operational perspective.
Key research facilities, infrastructure and equipment	Mature cloud FMIS; APIs/data interfaces; field, machinery and map-based farm management; system integrations; multidisciplinary agronomic/IT team. Proprietary customer-data access is not required.
Previous and current involvement in EU-funded research and training programmes/actions/projects	No EU-funded R&I project is relied upon. Applied R&D capacity is evidenced by national project 2023-1.1.1-PIACI-FÓKUSZ-2024-00022 on integrated agricultural data and decision support, completed in 2026.

6. Additional ethics information

FIELDWISE does not involve medical research, animals or sensitive-category human data. Human involvement is limited to voluntary grower/advisor/software-user feedback during demonstrations and the AgroVIR placement. Participants will receive clear information on purpose, data use and withdrawal; only data necessary for usability and operational assessment will be collected. Personal data will be minimised and, where applicable, coded/pseudonymised and processed under GDPR and institutional procedures. Commercially sensitive farm/customer information will be used only under agreed access and confidentiality conditions.

7. Additional information on security screening

Not applicable. FIELDWISE does not involve classified information, security-sensitive technologies, export-controlled materials or activities requiring additional security screening.

8. Environmental considerations in light of the MSCA Green Charter

FIELDWISE follows the MSCA Green Charter through data- and travel-efficient implementation. Existing five-season MATE data are reused before new measurements are generated; field campaigns are coordinated around critical physiological/Sentinel dates; and UAV flights are limited to supplementary observations when scientifically justified. Physical mobility is organised in scientifically necessary blocks, while routine coordination uses online meetings and public transport/rail is preferred where feasible. Digital-first dissemination, electronic documentation, equipment reuse and parsimonious/reproducible computational workflows further reduce environmental burden.

9. Required for Global Fellowships only: Letter of commitment from the associated partner hosting the outgoing phase

Not applicable – FIELDWISE is a European Postdoctoral Fellowship.

Use of generative artificial intelligence in proposal preparation

OpenAI ChatGPT was used in Parts B-1 and B-2 for language editing, structural refinement, consistency checking. The applicant critically reviewed and validated all AI-assisted content, scientific statements, methodology, references and final wording and retains full responsibility for the proposal. Generative AI was not used to fabricate empirical data or research results.



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